

- 3** The Indian Space Research Organisation planned to launch a geostationary satellite to provide broadcasting services. The mass of the satellite is 5.9×10^3 kg and the mass of the Earth is 5.9×10^{24} kg. The Earth has a radius of 6.4×10^6 m.

- (a)** State one advantage and one disadvantage of using a geostationary orbit for a broadcasting satellite.

Advantage

.....

Disadvantage

..... [2]

- (b)** Show that,

- (i)** the angular velocity of the geostationary satellite when it is in orbit around the Earth is $7.3 \times 10^{-5} \text{ rad s}^{-1}$.

[1]

- (ii)** the radius r of the geostationary orbit is 4.2×10^7 m.

[2]

- (c) Before deciding on what fuel can be used to propel the satellite into space, engineers need to determine the minimum amount of energy required to bring the satellite into the geostationary orbit.
- (i) Determine the kinetic energy of the satellite when it is in the geostationary orbit.

kinetic energy = J [3]

- (ii) Hence, determine the minimum total energy of the satellite required to launch the satellite from the surface of the Earth to its geostationary orbit. Assume the satellite begins its ascend with zero kinetic energy.

minimum total energy = J [3]

- (iii) Give a reason why the amount of energy required to bring the satellite into orbit should be more than that calculated in (c)(ii).

.....

..... [1]

[Turn over